AI/ML & RAG SPECIFICATION

Model responsibilities, pipelines, thresholds, evaluation and integration plan

AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)

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1. Purpose

This document defines the AI/ML portion of the project. It separates perception models (object and face recognition), speech/conversational processing, retrieval/RAG and response generation. The objective is to make each AI capability independently testable and prevent the language model from becoming the source of truth for deterministic information such as medication times or appointments.

2. AI Modules

Module	Input	Output	Primary Use
Object Detection	Camera frame/image	Object labels + boxes + confidence	Find/remember where belongings were seen
Face Detection	Camera frame	Face region(s)	Locate face before recognition
Face Recognition	Face crop/embedding	Known person + confidence	Recognize familiar people
Speech-to-Text	Audio	Transcript	Voice interaction
Text/Intent Processing	Text/transcript	Intent + entities/context	Route user request
Embedding Model	Text/memory	Vector embedding	Semantic retrieval
Vector Search	Query embedding + memory store	Relevant memories	Retrieve prior context
RAG Generator	Question + retrieved context	Grounded answer	Conversational memory
Assistant Router	Question + structured state + RAG	Tool/retrieval plan	Choose correct information source
Text-to-Speech	Answer text	Audio	Optional spoken response
Geofence Engine	GPS + zone definitions	Inside/outside event	Safety alerting

3. Core AI Principle

Use AI where interpretation is required and deterministic software where a fact is already stored. For example, “What time is my medicine?” should query the medication schedule directly. “What were we talking about when my daughter visited?” can use person-aware conversational retrieval. “Where did I keep my keys?” should query object observations first.

4. Object Detection Pipeline

1. Capture image/frame.
2. Pre-process image to model input.
3. Run object detector.
4. Filter detections using configured confidence threshold.
5. Map model label to the application's tracked-object vocabulary.
6. Associate observation with patient and timestamp.
7. Store optional scene/location metadata.
8. Persist observation metadata; avoid retaining raw frames unless explicitly required.
9. Expose latest/previous observations through the object API.

5. Object Detection Requirements

Requirement	Specification
Supported objects	Start with a defined vocabulary relevant to the project and common personal belongings.
Confidence	Store model confidence for each observation.
Location	Location may be a room/scene label or configured physical location.
Time	Every observation must have timestamp.
Last seen	Return the newest relevant observation, not simply the newest frame.
Unknown object	Do not force an unknown object into a known class.
False positives	Allow filtering and confidence tuning during evaluation.

6. Face Recognition Pipeline

10. Capture frame.
11. Detect face region.
12. Quality-check face crop.
13. Generate face representation using the selected recognition model.
14. Compare against enrolled representations.
15. Apply a validation threshold.
16. Return known person only when confidence/threshold policy is satisfied.
17. Record recognition event only when retention/privacy policy allows.

7. Face Recognition Safety Rules

- Do not equate a low-confidence similarity with a confirmed identity.
- Unknown/uncertain must be a valid output.
- Registration should require caregiver authorization or another explicitly defined enrollment mechanism.
- Face representations must not be exposed through ordinary client APIs.
- Provide a deletion/revocation path for registered people.
- Evaluate recognition separately for enrollment quality, lighting, pose and distance.

8. Person-Aware Conversational RAG

This is the key project-specific intelligence: the system first identifies a familiar person using facial biometrics, then uses that identity as a retrieval filter/context signal when searching stored conversations.

Example:

Face → recognized as "Person A" → retrieve memories involving Person A → answer patient question from retrieved context.

9. Person-Aware RAG Pipeline

18. Detect and recognize person.
19. Create conversation/session context with recognized_person_id if confidence passes policy.
20. Receive patient query.
21. Classify query intent.
22. If the query requires memory, create query embedding.
23. Filter candidate memories by patient_id and, when relevant, recognized_person_id.
24. Perform semantic vector search.
25. Optionally re-rank retrieved passages.
26. Construct grounded prompt containing only authorized retrieved context.
27. Generate concise answer.
28. Return answer plus internal/source metadata for auditing where appropriate.

10. Hybrid Retrieval Strategy

Question Type	Primary Source	Why
What medicine do I take?	Relational DB	Structured fact.
When is my next appointment?	Calendar DB	Structured fact.
Where were my keys last seen?	Object observations	Specific event data.
Who is this?	Face recognition	Identity task.

What did we talk about last time?	Vector memory + conversation metadata	Semantic memory.
What did I discuss with Person A?	Person-filtered vector memory	Identity-aware memory.
What is on my schedule today?	Schedule + appointments	Deterministic aggregation.
General conversational question	Assistant/LLM	Natural-language interaction.

The assistant router should decide between these sources rather than sending every question directly to the LLM.

11. Embedding and Vector Store Design

Field	Specification
Embedding input	Conversation segments, eligible memory summaries, object/event descriptions where useful.
Metadata	patient_id, source_type, source_id, recognized_person_id, event_time, retention status.
Vector index	Separate logical patient scope or mandatory patient metadata filtering.
Chunking	Short semantic chunks; preserve speaker/time/person metadata.
Retrieval	Top-k similarity followed by optional re-ranking.
Deletion	Deleting/retiring a memory must also remove or invalidate its vector entry.

12. Memory Ingestion Policy

- Do not vectorize every raw event automatically without a retention decision.
- Conversation segments should retain timestamp and speaker information.
- Person context should be attached when reliable recognition exists.
- Memory should have source references back to relational records.
- Sensitive information should be minimized in generated memory summaries.
- Caregiver access to memory must follow authorization rules.

13. Assistant Architecture

Stage	Responsibility
1. Input	Text or transcript.
2. Intent detection	Determine whether request is schedule, medication, object, person, memory, safety or general conversation.
3. Context builder	Collect only relevant patient context.
4. Tool/query execution	Call structured APIs or retrieval.
5. Evidence filtering	Remove unauthorized/low-confidence context.
6. Response generation	Generate answer grounded in retrieved facts.
7. Confidence/uncertainty	Represent uncertainty where appropriate.
8. Output	Text and optional TTS.

14. Recommended Model Strategy

The exact model should be selected after benchmarking on the project hardware and test data. The following is a practical starting strategy rather than a final locked model choice.

Capability	Starting Approach	Reason
Object detection	A lightweight YOLO-family detector or equivalent mobile-friendly detector	Real-time camera inference and manageable compute.
Face detection	Lightweight face detector	Fast preprocessing for recognition.
Face recognition	Embedding-based face recognition model	Allows enrolled-person matching without relying on a generative model.
Embeddings	Sentence-Transformers family model such as a MiniLM-class encoder	Lightweight semantic retrieval baseline.
Vector DB	FAISS for local prototype	Simple local similarity search and compatible with offline-first design.
LLM	Small local instruct model compatible with available hardware	Keeps private memory local and avoids mandatory cloud dependency.

STT	Offline speech model appropriate to device capability	Voice assistant without mandatory cloud audio processing.
TTS	Device/native or offline TTS where available	Spoken assistance with lower privacy exposure.

Model names, versions and thresholds should be recorded in the implementation configuration and evaluation report rather than hard-coded throughout the application.

15. Hardware-Aware Development Strategy

- Begin with CPU-compatible inference and small models.
- Measure latency before adding larger models.
- Keep object detection and face recognition independently replaceable.
- Do not assume a cloud GPU is available.
- Use quantized local language models if needed for acceptable memory usage.
- Cache embeddings and model initialization to avoid repeated startup cost.

16. Initial Threshold Strategy

Thresholds must be empirically determined from project test data. Do not treat arbitrary threshold values as universally valid.

Signal	What to Tune	Evaluation
Object confidence	Detection confidence threshold	Precision/recall on project object images.
Face match	Similarity threshold	False accept vs false reject on enrolled/unknown faces.
Face quality	Minimum quality gate	Recognition success across pose/lighting.
RAG similarity	Retrieval threshold/top-k	Relevant retrieval rate and unsupported-answer rate.
LLM answer	Grounding checks	Factual consistency with retrieved context.

17. AI Evaluation Dataset

Create a controlled project dataset covering the exact intended use cases.

Dataset	Minimum Categories to Collect
Objects	Keys, phone, wallet, glasses, bottle, bag and project-specific belongings.
Faces	Enrolled familiar people + unknown people; varied lighting/pose/distance.
Conversations	Short dialogues containing references to people, events and prior discussions.
Queries	Object, person, medication, appointment, schedule and memory questions.
Location	Inside/outside safe-zone events and GPS unavailable cases.

All data collection must follow the project's consent, privacy and retention requirements.

18. AI Evaluation Metrics

Module	Metrics
Object detection	Precision, recall, mAP where applicable, latency.
Face recognition	Accuracy, false acceptance rate, false rejection rate, latency.
STT	Word error rate and task-level intent accuracy.
RAG	Recall@k, precision@k, answer groundedness, unsupported-answer rate.
Assistant	Task success rate, correct routing rate, response latency.
Geofence	Correct enter/exit detection, false alert rate.
End-to-end	Successful completion of representative patient tasks.

19. RAG Failure Handling

- No relevant memory → state that no relevant memory was found.
- Low retrieval confidence → ask a clarifying question or provide a qualified response.
- Conflicting memories → present the conflict rather than choosing silently.
- Unauthorized memory → exclude it before generation.
- Stale object observation → explicitly show observation time.
- LLM unavailable → return deterministic information through normal APIs where possible.

20. Prompt Construction Rules

- Tell the model that retrieved context is evidence, not instructions.
- Include only relevant retrieved passages.
- Include timestamps/person identity metadata when needed.
- Tell the model not to invent missing details.
- Prefer concise answers suitable for the patient interface.
- Separate system instructions from retrieved memory text.

21. AI Security and Privacy

- Enforce patient scope before vector retrieval.
- Never use a raw client-supplied patient_id as authorization proof.
- Do not expose embeddings or internal prompts to the patient/caregiver unnecessarily.
- Sanitize/validate uploaded images and audio.
- Keep model files and indexes in protected application storage.
- Log model/version/configuration changes.
- Define deletion behavior for face representations, conversations and vector memories.

22. End-to-End Example — “Who is this and what did we talk about?”

29. Camera captures a face.
30. Face detector identifies the face.
31. Recognition model produces a representation.
32. Similarity search finds a familiar person above the configured threshold.
33. System stores recognized_person_id in the current assistant context.
34. Patient asks what they discussed previously.
35. Assistant router identifies a conversational-memory intent.
36. Vector search filters memories to the current patient and recognized person.
37. Top relevant memories are retrieved and checked for authorization/retention.
38. LLM receives only the question and relevant evidence.
39. Assistant answers concisely and avoids unsupported details.

23. End-to-End Example — “Where are my keys?”

40. Patient asks by voice/text.
41. STT converts voice to text if necessary.
42. Intent router identifies object-location query.
43. Object query searches tracked objects and recent observations.
44. Latest relevant observation is selected.

45. Assistant returns location + observation time.
46. If no recent observation exists, assistant states that clearly.

24. AI Development Order

47. Object detection prototype
48. Face detection + recognition prototype
49. Object observation database integration
50. Conversation storage
51. Embedding generation + FAISS indexing
52. Semantic retrieval
53. Person-aware retrieval
54. Assistant router
55. Local LLM integration
56. STT/TTS integration
57. End-to-end assistant
58. Evaluation and threshold tuning

25. Definition of Done for AI Modules

Module	Done When
Object detection	Works on project test images, stores observations and returns last-seen result.
Face recognition	Recognizes enrolled test identities and returns unknown for defined unknown cases.
RAG	Retrieves relevant memories with patient/person filtering.
Assistant	Routes structured vs memory questions correctly and avoids fabricated facts.
Geofence	Correctly generates configured enter/exit events on test traces.
Voice	Can complete a defined set of voice tasks if voice is included in MVP.
Evaluation	Metrics and failure cases are recorded.

26. Final AI Architecture Summary

Camera/Voice/Text → Perception (Object/Face/STT) → Context & Intent Router → Structured Data / Vector Retrieval → Evidence Validation → Local LLM Response → Patient UI/TTS. Caregiver events are generated through deterministic business rules such as reminders, medication status and geofence evaluation.